The empirical recurrence for OEIS A306132, recovered from its tail

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Abstract

OEIS A306132 records "Empirical recurrence of order 71" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{71} = -7680 \neq 0$, so no further tail satisfies anything shorter; any order-71 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A306132 is "Number of nX4 0..1 arrays with every element unequal to 0, 2, 3, 4, 6 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$1, 24, 64, 498, 2922, 16877, 111073, 700859, \ \ldots$$

The entry states:

> Empirical recurrence of order 71 (see link above)

As of the "Last modified" line on the live entry (Jun 22 2018 07:42:21, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 71: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 71.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 71 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{71} c_i \, a(n-i),$$

c_1	8	c_{19}	−907094	c_{37}	−1973907526	c_{55}	−79294966
c_2	−5	c_{20}	576325	c_{38}	2687570633	c_{56}	164509120
c_3	9	c_{21}	−1147815	c_{39}	−560421397	c_{57}	230710678
c_4	−358	c_{22}	−1150912	c_{40}	−263785901	c_{58}	−45298430
c_5	289	c_{23}	−7348770	c_{41}	3955632792	c_{59}	−213074994
c_6	1114	c_{24}	−3109858	c_{42}	−2517312415	c_{60}	−47527630
c_7	3444	c_{25}	7130684	c_{43}	302700042	c_{61}	94671854
c_8	−3929	c_{26}	−15294180	c_{44}	1885648125	c_{62}	24920174
c_9	−14259	c_{27}	34302105	c_{45}	−988662812	c_{63}	−32465683
c_{10}	−15343	c_{28}	49876443	c_{46}	−2246381701	c_{64}	−4978808
c_{11}	20918	c_{29}	−201681857	c_{47}	1121471425	c_{65}	8986704
c_{12}	45055	c_{30}	37545649	c_{48}	−1244087973	c_{66}	394928
c_{13}	129920	c_{31}	105335530	c_{49}	−1788998269	c_{67}	−2057260
c_{14}	−78869	c_{32}	−82123665	c_{50}	2075196777	c_{68}	−253520
c_{15}	−474001	c_{33}	344005356	c_{51}	74906918	c_{69}	227264
c_{16}	12942	c_{34}	−291145960	c_{52}	−1060060489	c_{70}	35104
c_{17}	490385	c_{35}	601920132	c_{53}	−71234841	c_{71}	−7680
c_{18}	1465962	c_{36}	461249562	c_{54}	667905205		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A306132 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{71} - c_1 x^{71-1} - \cdots - c_{71}$. Its constant term is $-c_{71} = 7680$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 71 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 71, so $\deg q = 71$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 71, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 71, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -7680.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-71 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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